

Question 1 (12 Marks)

(a) Simplify $2\sqrt{8} - \sqrt{18}$.

(b) If $\frac{\sqrt{3}-4}{2+\sqrt{3}} = a+b\sqrt{3}$, find a and b .

(c) Simplify $\frac{x^2-2x-3}{x^2-4x-5} \times \frac{x^2-25}{(x-3)(x+5)}$.

(d) Find the simultaneous solution to the equations: $2x = 3y + 1$ and $xy + x + y = 23$.

Question 2 (12 Marks) *Start a new page*

(a) Factorise

(i) $y^3 - 125$.

(ii) $1 + x + x^2 + x^3$.

(b) Solve

(i) $2x^2 = 7x - 5$.

(ii) $\frac{5x+3}{x+4} > 2$.

(iii) $|4x+3| \geq 1$.

(c) Given $f(x) = 2(x^3 - x)$

(i) Find $f(-x)$.

(ii) Show whether $f(x)$ is odd, even or neither.Question 3 (12 Marks) *Start a new page*

(a) Draw a neat sketch of each of the graphs below.

(i) $y = \frac{1}{x}$.

(ii) $y = \frac{1}{x-3}$.

(iii) $y = 4 + \frac{1}{x-3}$.

(b) (i) Find the domain and range for the function $f(x) = \sqrt{9-x^2}$.(iii) Sketch the region indicated by $y \geq \sqrt{9-x^2}$.

(c) Solve $|x+1| - |2x-1| < 0$.

End of Paper

Question 1

$$\begin{aligned} \text{(a)} \quad 2\sqrt{8} - \sqrt{18} \\ = 4\sqrt{2} - 3\sqrt{2} \\ = \sqrt{2} \end{aligned}$$

$$\text{(b)} \quad \frac{\sqrt{3}-4}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}}$$

$$= \frac{(\sqrt{3}-4)(2-\sqrt{3})}{(2+\sqrt{3})(2-\sqrt{3})}$$

$$= \frac{\sqrt{3}-8-3+4\sqrt{3}}{4-3}$$

$$= 6\sqrt{3}-11 = -11+6\sqrt{3}$$

$\therefore a = -11, b = 6$

$$\text{(c)} \quad \frac{x^2-2x-3}{x^2-4x-5} \times \frac{x^2-25}{(x-3)(x+5)}$$

$$= \frac{(x-3)(x+1)}{(x-5)(x+1)} \times \frac{(x-5)(x+5)}{(x-3)(x+5)}$$

$$= 1$$

$$\text{(d)} \quad 2x = 3y+1 \quad \text{--- ①}$$

$$xy + x + y = 23 \quad \text{--- ②}$$

rearrange ①

$$x = \frac{3y+1}{2} \quad \text{--- ③}$$

sub ③ in ②

$$\left(\frac{3y+1}{2}\right) \times y + \left(\frac{3y+1}{2}\right) + y = 23$$

$$y(3y+1) + (3y+1) + 2y = 46$$

$$3y^2 + y + 3y + 1 + 2y = 46$$

$$3y^2 + 6y - 45 = 0$$

$$y^2 + 2y - 15 = 0$$

$$(y+5)(y-3) = 0$$

(d) $\therefore y = -5$ or 3

Sub y in ①

when $y = -5$ $2x = -15 + 1$

$$2x = -14$$

$$\therefore x = -7$$

when $y = 3$ $2x = 9 + 1$

$$2x = 10$$

$$\therefore x = 5$$

Check in ②

$$-7x - 5 + -7 + 5 = 23 \checkmark$$

$$5 \times 3 + 5 + 3 = 23 \checkmark$$

$\therefore$ the solutions are when $x = -7, y = -5$ }
and $x = 5, y = 3$ }

Question 2

(a) (i) $y^3 - 125 = (y)^3 - (5)^3$
 $= (y - 5)(y^2 + 5y + 25)$

(ii) $1 + x + x^2 + x^3$
 $= 1(1 + x) + x^2(1 + x)$
 $= (1 + x)(1 + x^2)$

(b) (i) Solve $2x^2 = 7x - 5$

$$2x^2 - 7x + 5 = 0$$

$$2x^2 - 2x - 5x + 5 = 0$$

$$2x(x - 1) - 5(x - 1) = 0$$

$$(x - 1)(2x - 5) = 0$$

$$\therefore x = 1 \text{ or } \frac{5}{2}$$

P	S
$a \times b = 10$	$a + b = -7$
$a = -5$	$b = -2$

(ii) $\frac{5x + 3}{x + 4} > 2$

$$(x + 4)^2 \times \frac{5x + 3}{x + 4} > 2(x + 4)^2$$

$$(x + 4)(5x + 3) > 2(x + 4)^2$$

Question 2 continued

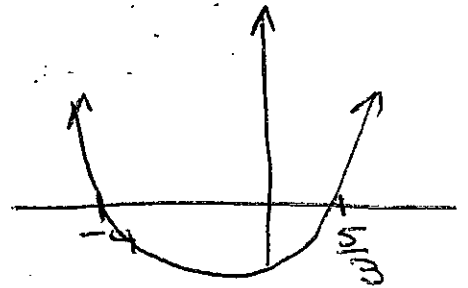
$$(x+4)(5x+3) - 2(x+4)^2 > 0$$

$$(x+4)(5x+3-2(x+4)) > 0$$

$$(x+4)(5x+3-2x-8) > 0$$

$$(x+4)(3x-5) > 0$$

$$x = -4, \frac{5}{3}$$



∴ $x < -4$ or $x > \frac{5}{3}$

(iii) $|4x+3| \geq 1$

$$4x+3 \geq 1 \quad \text{or} \quad 4x+3 \leq -1$$

$$4x \geq -2 \quad \text{or} \quad 4x \leq -4$$

$$\therefore x \geq -\frac{1}{2} \quad \text{or} \quad x \leq -1$$

(c) $f(x) = 2(x^3 - x)$

(i) $f(-x) = 2((-x)^3 - (-x))$
 $= 2(-x^3 + x)$

(ii) $f(-x) = 2(-x^3 + x)$
 $= -2(x^3 - x)$

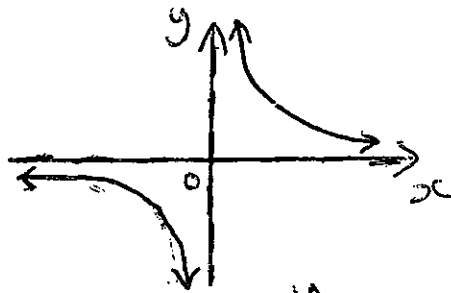
$$= -f(x) = -1(2(x^3 - x))$$

$$= -2(x^3 - x)$$

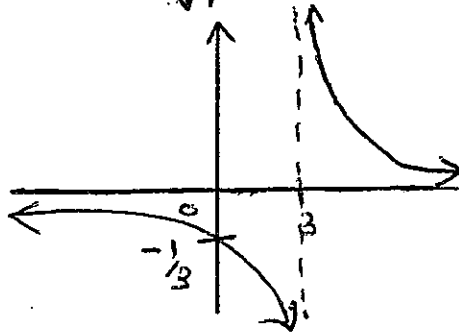
$$\therefore f(-x) = -f(x)$$

∴ $f(x)$ is an odd function.

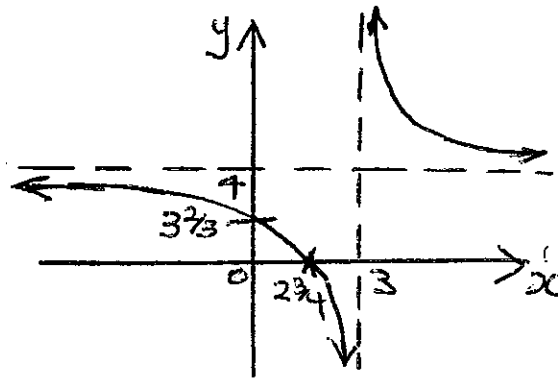
Question 3
a) (i) $y = \frac{1}{x}$



(ii) $y = \frac{1}{x-3}$



(iii) $y = 4 + \frac{1}{x-3}$



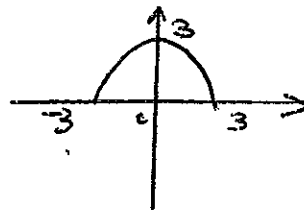
(b) $f(x) = \sqrt{9-x^2}$

(i)

Condition $9-x^2 \geq 0$

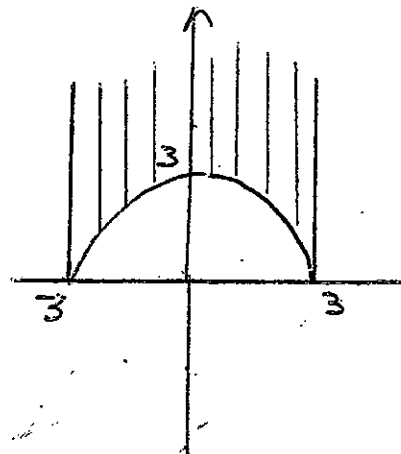
$(3-x)(3+x) \geq 0$

$\therefore$ Domain $= -3 \leq x \leq 3$



Range: $0 \leq y \leq 3$

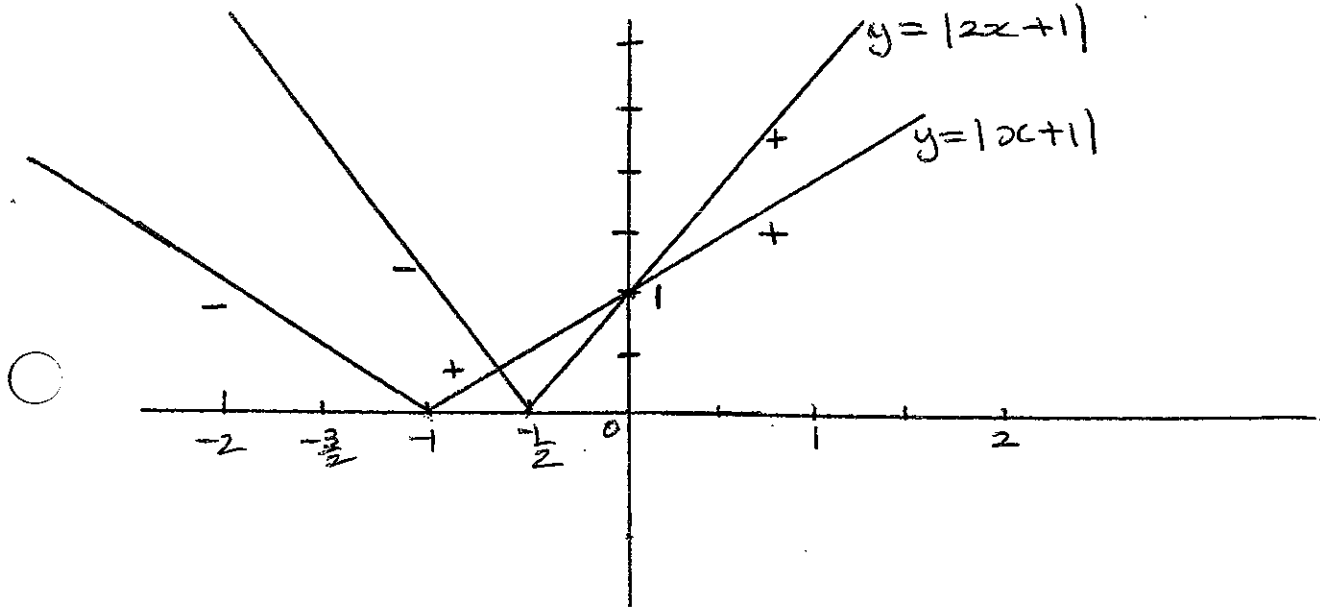
(ii) $y \geq \sqrt{9-x^2}$



Question 3 continued

(c) $|x+1| - |2x-1| < 0$
 $|x+1| < |2x+1|$

graph $y = |x+1|$ and $y = |2x+1|$



points of intersection

$$x+1 = 2x+1$$

$$0 = x$$

$$x+1 = -(2x+1)$$

$$x+1 = -2x-1$$

$$3x = -2$$

$$x = -\frac{2}{3}$$

the solution is

$$x > 0 \text{ or } x < -\frac{2}{3}$$